Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

Level 0 - Context Diagram

```
          Free-Text Input

     Student                                        Emotion Detector Platform

     Detected Emotion + Guidance

                                                     API Calls


                                                      Google Gemini API
```

Level 1 - High-Level Data Flow

```
Student Input (Free Text)



   1. Preprocessing     Clean text, remove noise, tokenize




  BiLSTM      BERT      (Parallel inference)
  Model      Model



         Probability Vectors


   2. Ensemble Engine    40% BiLSTM + 60% BERT weighted average



   3. Rule-Based Boost      Keyword matching  confidence boost (+15%)
```

Level 2 - Detailed Data Flows

2.1 Preprocessing Flow

Input	Process	Output
Raw student text	Remove HTML, URLs, mentions	Cleaned text
Cleaned text	Lowercase, remove punctuation	Normalized text
Normalized text	NLTK stopword removal	Filtered tokens
Filtered tokens	WordNet lemmatization	Final tokens
Tokens (BiLSTM path)	Integer encoding + padding (maxlen=100)	Padded sequence
Tokens (BERT path)	BertTokenizer (bert-base-uncased)	input_ids + attention_mask

2.2 Model Inference Flow

Model	Input	Output
BiLSTM	Padded integer sequence (100,)	Softmax probability vector (5,)
BERT	input_ids + attention_mask	Softmax probability vector (5,)
Ensemble	2 probability vectors	Weighted average vector (5,)

2.3 Gemini Guidance Flow

Input	Process	Output
Final emotion + confidence	Prompt builder emotion-specific template	Structured prompt
Structured prompt	Gemini API call (gemini-pro)	5-section guidance text
Guidance text	HTML formatting	Rendered guidance panel

Data Entities

Entity	Description	Storage
Student Input	Raw text describing study problem	Streamlit session state
Detection Result	Emotion, confidence, probabilities,	Streamlit session state + CSV
Guidance Text	5-section AI-generated learning support	Streamlit session state
Interaction Log	Full row per session	data/interactions/interaction_logs.csv
Model Weights	BiLSTM (.h5) and BERT (safetensors)	data/models/ (local only)